

Shreya Chakraborty

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Education

Masters in Artificial Intelligence and Business Analytics, University of South Florida, Tampa, FL

CGPA 3.81 May 2025

Relevant Coursework: **Statistical Data Mining, Text Analytics, Big Data, Distributed Information Systems**

Data Science Programming, Database Management Systems, Advanced System Design

Enterprise Information Systems, Data Mining, Analytical Methods for Business

Bachelors in Electronics and Communication Engineering, MIT, Manipal, India

Work Experience

Data Science Program - USF

Feb 2024 – Present

- Developed Python and ML scripts to automate certificate issuance and detect duplicate records, streamlining administrative processes. providing 100% improvement in accuracy and 40% improvement in total process time
- Maintained a master SQL database for the course, ensuring organized and accurate data management.
- Served as a Student Mentor at the University of South Florida for the Citizen Data Science Certificate program in collaboration with Tableau; delivered presentations and guided undergraduates in building interactive dashboards and data-driven stories for projects such as consumer sentiment analysis for Cigar City Brewing (75,000+ BeerAdvocate reviews) and stock market forecasting using regression models and statistical indicators.

Data Engineer/Analyst – Accenture India

May 2021 – May 2023

- Built ETL logic to extract client-provided Excel data, transform and validate it, load it into an SQL database, and then used Python and Automation Anywhere to automate the creation of 100,000 sales and acquisition forms within 48 hours.
- Built an ETL pipeline orchestrated with Airflow DAGs to extract FRED economic data, preprocess and cache it using Python pickle files, and load it into Oracle DB using stored procedures with dynamic bookmark logic to prevent duplication.
- Developed a Flask web app using the MovieLens dataset and Oracle DB to deliver personalized movie recommendations via cosine similarity, with stored procedures for rating insertion and secure authentication.
- Received ‘Sparkling Star Award’ and client appreciation at Accenture in 2022

Project Intern - Bizz4Solutions, India

Dec 2020- May 2021

- Implemented algorithms using python to predict revision time intervals personalized for a student based on his test score history
- Test results of 10,000 students over 12 test cycles were used to build the algorithms. Prediction accuracy was 100% on a sample size of 100 students. To manage the large dataset (2 TB), Mongo DB in AWS environment was used

Projects Experience

May 2024- May 2025

Predicting the success of bank telemarketing using data modelling

- Developed a machine learning model to predict bank telemarketing success, analysing 45,000 client records
- Applied AI/ML models **XGBoost, SVM, Random Forest, Neural Networks, Logistic Regression**, and **Decision Tree**, used **hyperparameter tuning** to improve performance. **Logistic Regression achieved the highest recall score of 93%**, ensuring effective identification of potential subscribers

Big Data Processing & Predictive Analytics using PySpark

- Study of Racial Bias in Police Stops Analyzed 3.2M+ police stop records from the Stanford Open Policing Project using PySpark for data ingestion, cleaning, and transformation. Addressed missing values, standardized column names, and stored the processed data in a persistent table
- Conducted Exploratory Data Analysis (EDA) to examine arrests by race, gender, and age. Identified key patterns, including racial disparities in stop rates and a moderate correlation between age and arrest likelihood.
- Tackled big data challenges such as memory management in a distributed environment while handling large datasets in Spark.
- Developed insights that will contribute to a binary prediction model using Spark ML for identifying potential biases in policing **outcomes**.

Statistical Analysis on Evolving Patterns of Drug Overdose-Related Mortality Rates

- Conducted a nationwide analysis of average drug overdose incidents across all U.S. states, evaluating the impact of diverse state-level policy.
- Collected, cleaned, and integrated multi-source data from government databases, ensuring high data integrity for reliable insights
- Utilized R to apply Linear Mixed-Effects models and Panel Data Analysis for evaluating policy effectiveness over 8 years across 50 U.S. states.
- Modelled hierarchical structures for time and geography using log-transformed data, and compared pooled, fixed, and random effects models

AI-Interview Agent System

- Designed and implemented an AI-powered interview agent using **LangChain** that automates job posting, custom question generation, candidate response collection, and scoring using structured JSON-based workflows and RESTful API endpoints.
- Built modular backend architecture with endpoints for posting job descriptions, creating role-specific questions, retrieving interview forms, submitting responses, and rating candidates ensuring a seamless flow between recruiter and candidate interactions.
- Engineered persistent data storage system using uniquely identified JSON files, allowing easy tracking of job roles, candidate responses, and final ratings for informed hiring decisions.

Media Bias Project

- Scrapped data from allsides website using selenium
- Used BERT Transformer from the Hugging Face Library to classify Newspaper Articles as Left, Right or Centre

Weather App Using ASP.NET

- Developed a dynamic Weather Dashboard using **ASP.NET MVC**, integrating WeatherAPI.com to display real-time and historical weather data with interactive Chart.js visualizations.
- Deployed the application on Microsoft Azure, ensuring global accessibility; utilized **C#, HTML, CSS, Bootstrap** for a responsive UI.
- Link to Website: <https://weatherappshr.azurewebsites.net/>

Skills

Cloud Platforms - AWS (S3, Lambda, DynamoDB, Athena, SQS, EC2, RDS), Azure

AI/ML - Deep Learning Models (Transformers, CNNs, RNNs), Self-Supervised Learning, Zero-shot Models, Prompt Engineering, Hugging Face Transformers

Frameworks - Spring Boot, Django, Flask, Entity Framework, ASP.NET

Programming Languages – Python, Java, C#, C++, SQL, HTML, JavaScript

Big Data Technology – Mongo DB, Hadoop, Spark, MapReduce, Apache Airflow, Snowflake, Kubernetes, Containerization, CI/CD, Shell Scripting

Version Control and Database – GitHub, Git, SQL Server, PostgreSQL, Cassandra